1 BFS pseudocode

```
function BFS(graph, sourceVertex) {
  for (u in graph.vertices) {
    u.color = WHITE
    u.d = infinity
    u.p = NIL
  }
  sourceVertex.color = GRAY
  sourceVertex.d = 0
  Q = empty queue
  enqueue(Q, sourceVertex)

  while (!isEmpty(Q)) {
    u = dequeue(Q)
    for (neighbor in graph.adjacencyList[u]) {
      if (neighbor.color == WHITE) {
        neighbor.color = GRAY
        neighbor.d = u.d + 1
        neighbor.p = u
        enqueue(Q, neighbor)
      }
    }
    u.color = BLACK
  }
}
```

2 DFS pseudocode

```
function DFS(graph) {
  global time = 0 // global variable

  for (u in graph.vertices) {
    u.color = WHITE
    u.p = NIL
  }

  for (u in graph.vertices) {
    if (u.color == WHITE) {
      DFSVisit(G, u)
    }
  }
}

function DFSVisit(graph, u) {
```

```
  time++
  u.d = time
  u.color = GRAY

  for (v in graph.adjacencyList[u]) {
    if (v.color == WHITE) {
      v.p = u
      DFSVisit(graph, v)
    }
  }

  u.color = BLACK
  time++
  u.f = time
}
```